

Initial diagnostic and essential commands

# Bioinformatic tools

Corentin Maslard

September 14, 2026

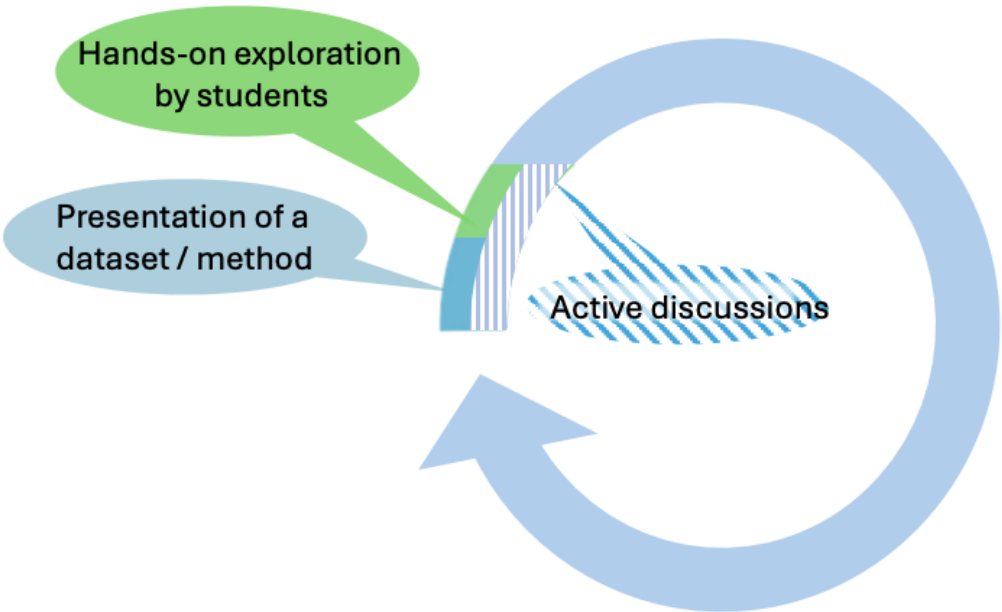
# Course organisation

## Bioinformatic tools

MSc Biology, 2026/2027 Autumn semester

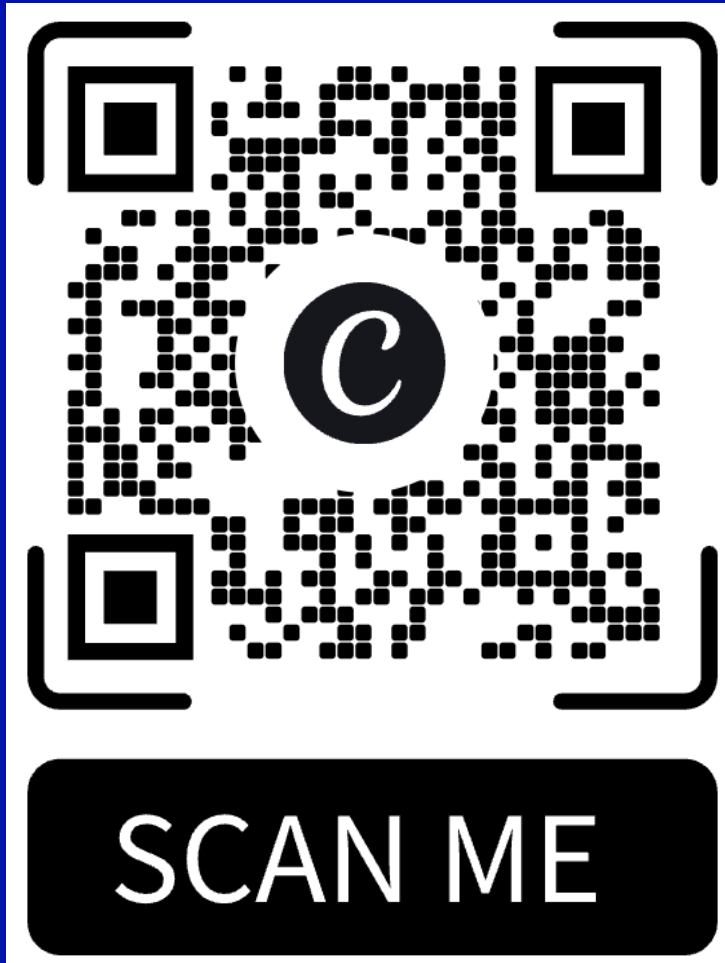
Maslard Corentin ([corentin.maslard@unine.ch](mailto:corentin.maslard@unine.ch), office A313)

|           | Monday              | Tuesday | Wednesday | Thursday | Friday |
|-----------|---------------------|---------|-----------|----------|--------|
| 8h15-9h   |                     |         |           |          |        |
| 9h15-10h  |                     |         |           |          |        |
| 10h15-11h |                     |         |           |          |        |
| 11h15-12h |                     |         |           |          |        |
| 13h15-14h |                     |         |           |          |        |
| 14h15-15h | Bioinformatic tools |         |           |          |        |
| 15h15-16h |                     |         |           |          |        |
| 16h15-17h |                     |         |           |          |        |
|           |                     |         |           |          |        |



**!! Daily participation and feedback required**

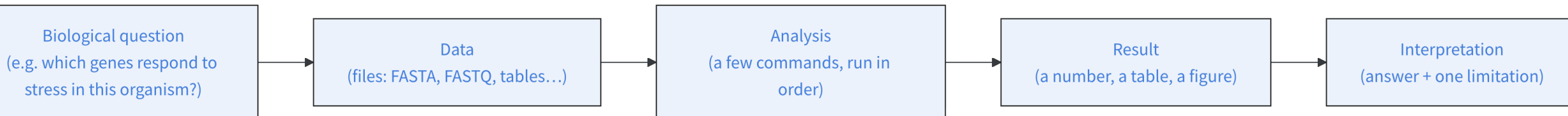
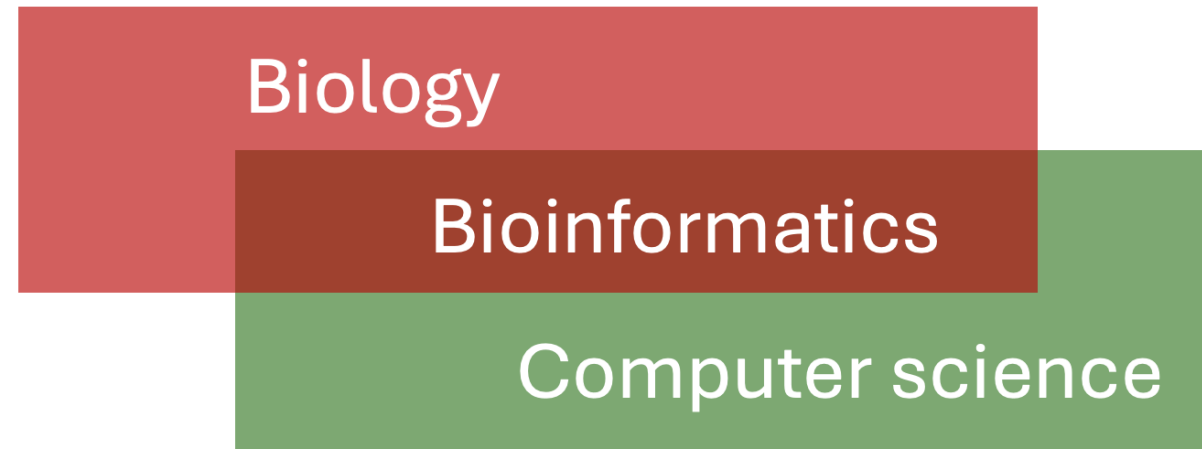
- > Continuous participation [30]
- > Course exercises [70]
- 10 questions related to the hands-on exercise at each of the seven courses /  
See guidelines / info on the **moodle** page (Soon)

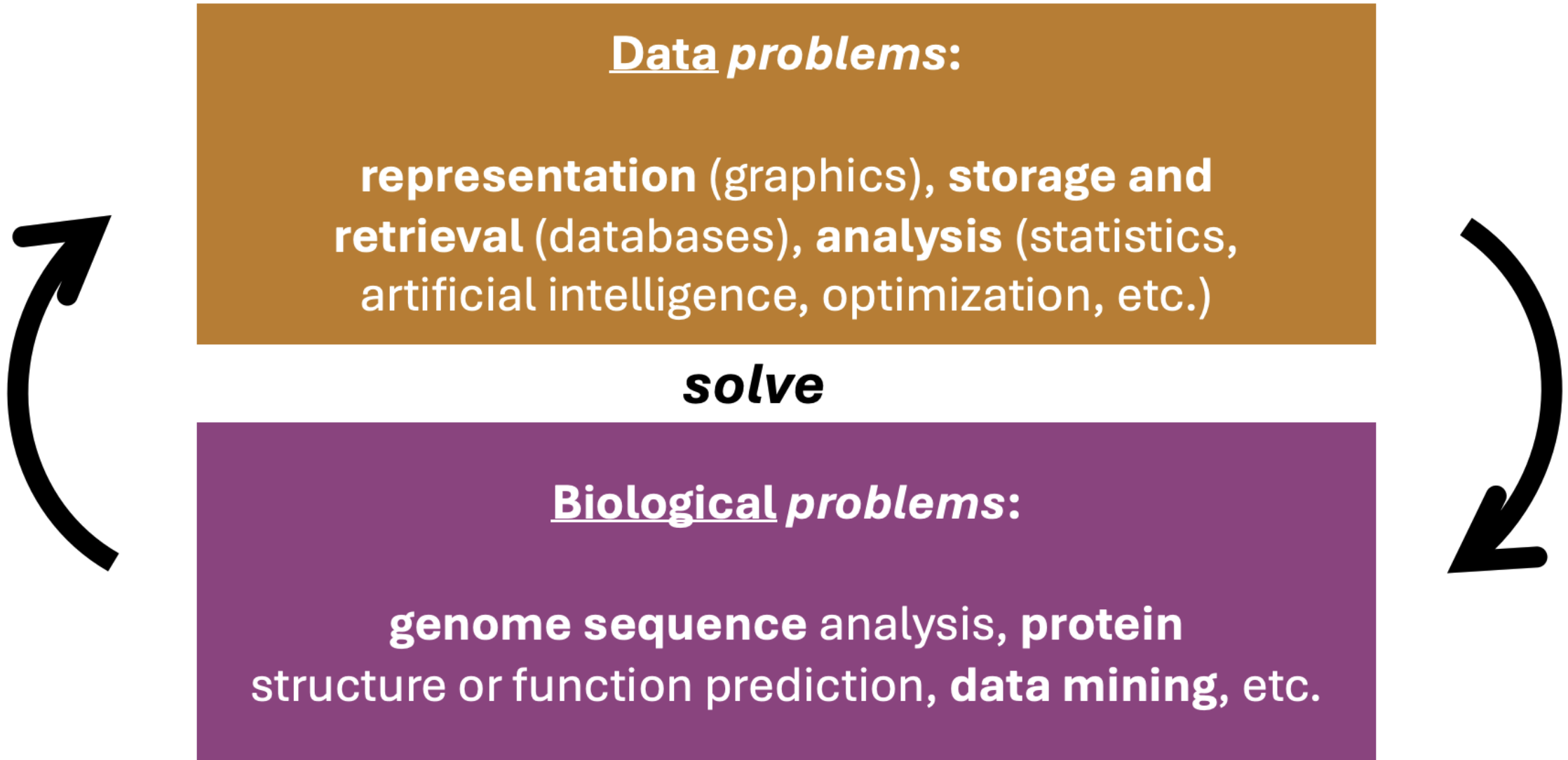


## Starting-point survey

# What is bioinformatics?

"Bioinformatics is the field of science in which biology, computer science, and information technology merge into a single discipline." *NCBI, 2001*

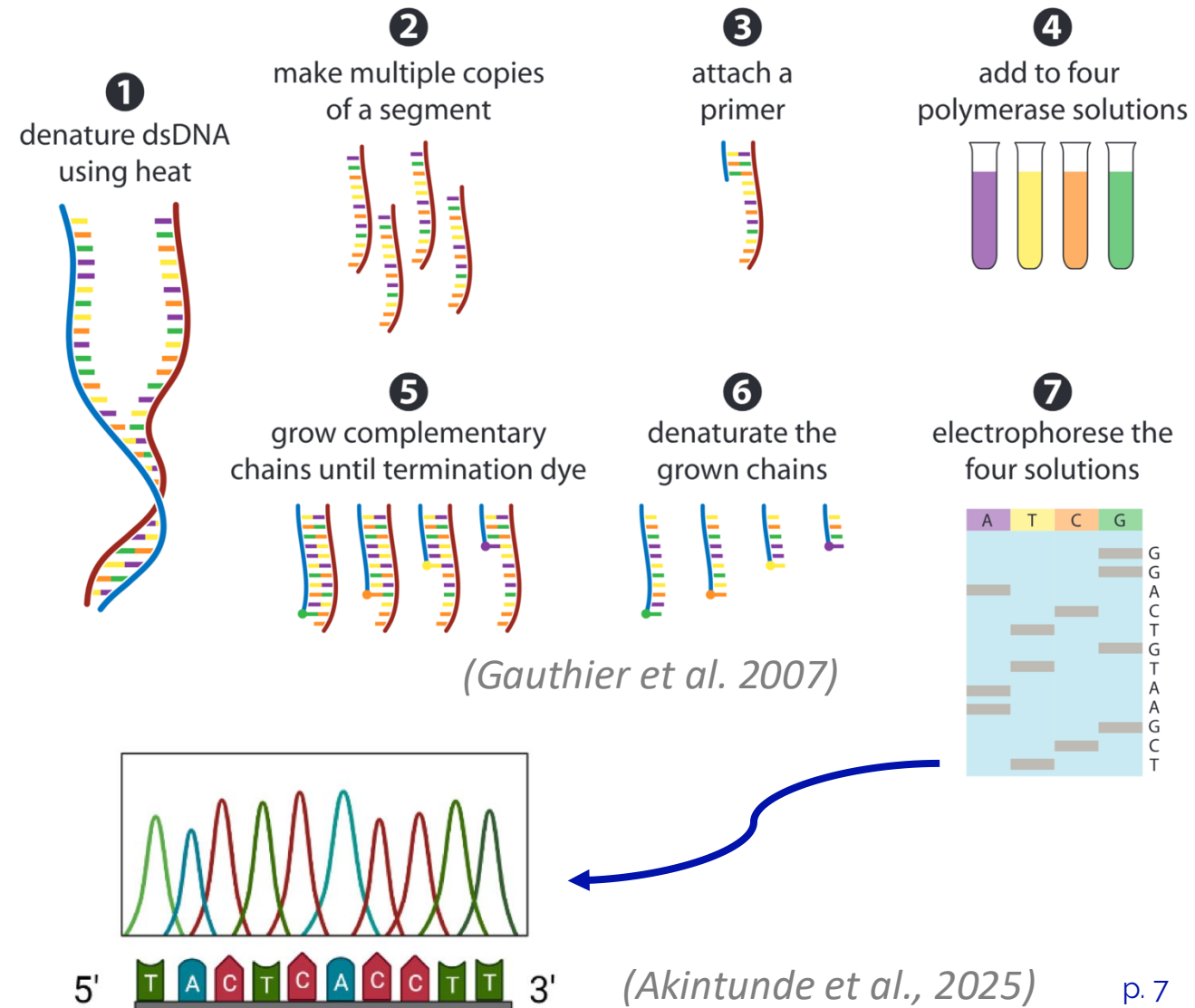




# The Sanger sequencing method in 7 steps.

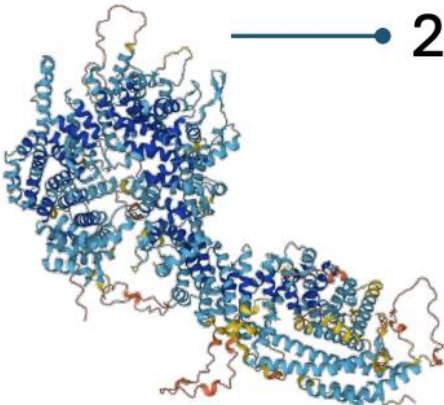
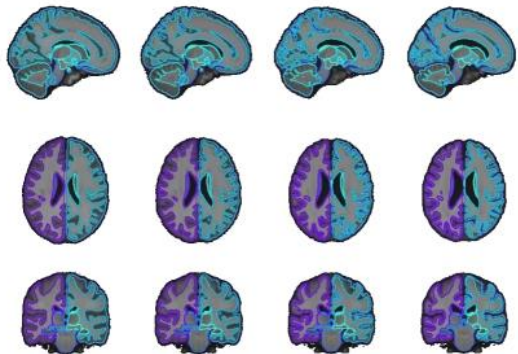
- 1977: First way to read DNA (Sanger sequencing)

- (1) The dsDNA fragment is denatured into two ssDNA fragments.
- (2) A fragment of ssDNA is multiplied into millions of copies.
- (3) A primer that corresponds to one end of the fragment is attached.
- (4) The fragments are added to four polymerase solutions. Each solution contains the four types of bases but only one type of termination nucleotide.
- (5) The chain grows until a termination nucleotide is randomly added.
- (6) The resulting dsDNA fragments are denatured to obtain a series of ssDNA of various lengths.
- (7) The fragments are separated by electrophoresis and the sequence is read.



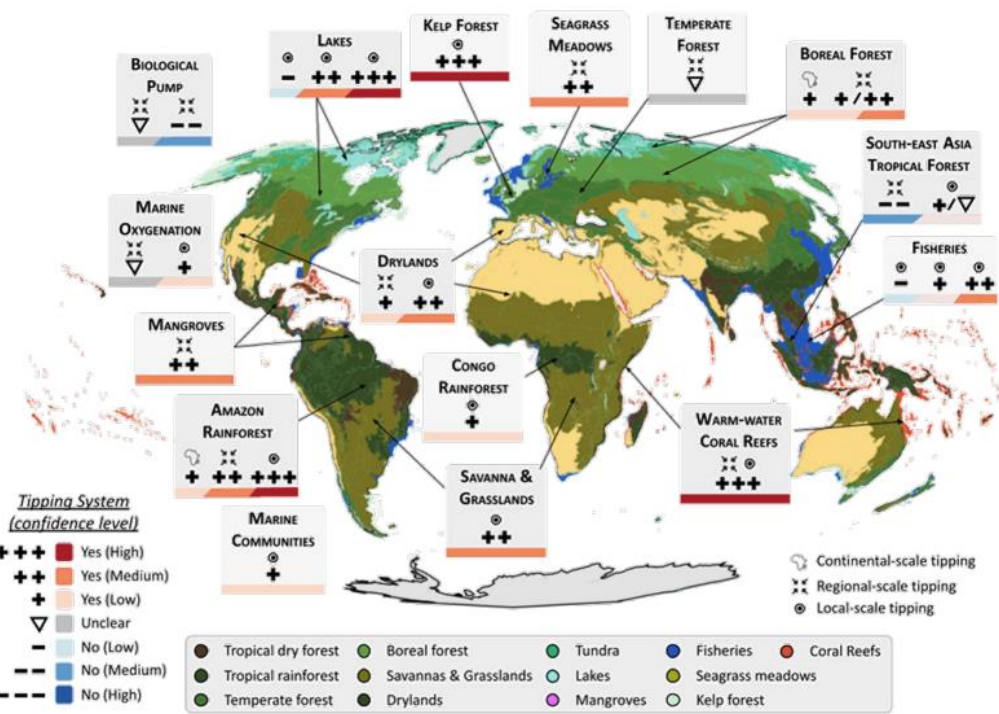
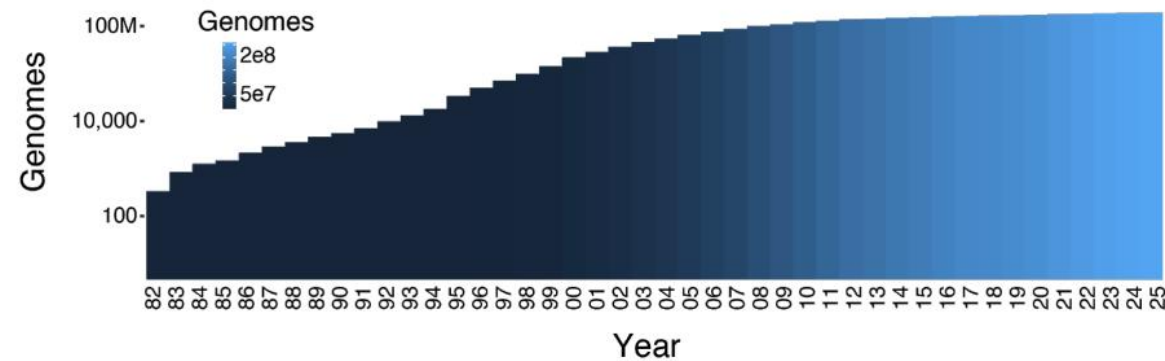
# Large data everywhere

Billions+ of images



200M+ protein structures

100M+ assembled genomes



Billions of bioclimatic data points



## A young field, a few landmarks

- 1977: First way to read DNA (Sanger sequencing)
- 1990: BLAST, compare a sequence to a whole database (→ Session 2 or 3)

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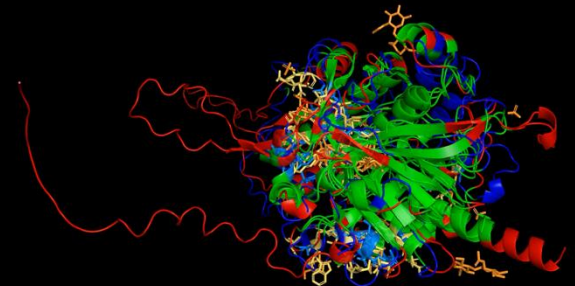
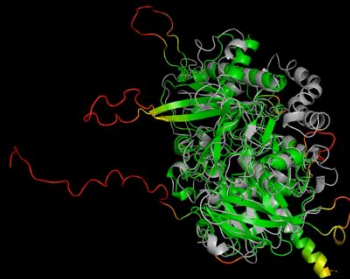
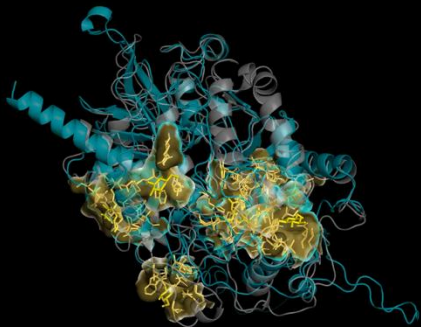
- 1977: First way to read DNA (Sanger sequencing)
- 1990: BLAST, compare a sequence to a whole database (→ Session 2 or 3)
- 2001: First draft of the human genome

13 years and \$3 billion in 2001  
Today, one day and a few hundred dollars



## A young field, a few landmarks

- 1977: First way to read DNA (Sanger sequencing)
- 1990: BLAST, compare a sequence to a whole database (→ Session 2 or 3)
- 2001: First draft of the human genome
- 2007-2008: High-throughput sequencing: data explodes
- 2020: [AlphaFold](#): deep learning meets protein structure (→ Session 7 ?)



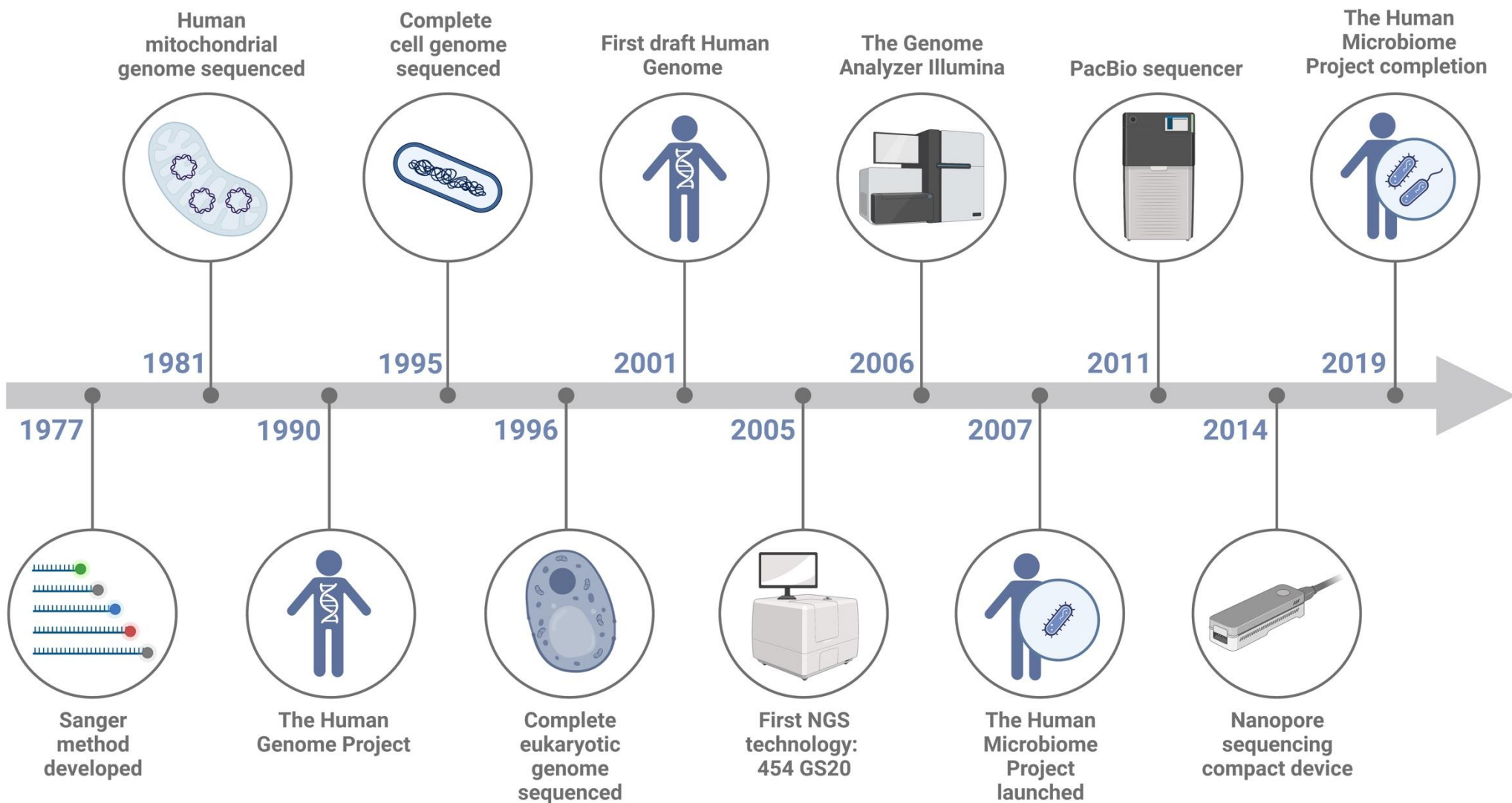
## A young field, a few landmarks

- 1977: First way to read DNA (Sanger sequencing)
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You will touch a small piece of each of these ideas in this course



# The History of DNA Sequencing



# Where does biological data come from?

## Different instruments, different questions.

- Genome/amplicon sequencing → FASTA, FASTQ
- RNA-seq (bulk or single-cell) → count tables
- Proteomics / metabolomics → abundance tables
- Imaging → measurement tables
- Genotyping / GWAS → genotype tables

Whatever the instrument: it almost always ends up as  
**sequences or tables.**

## How a computer works

- RAM: working memory, erased when powered off
- Disk/SSD: storage, kept when powered off

# Units of digital information

From bit to petabyte (decimal and binary units, with English and French abbreviations)

1 byte (B) = 8 bits

Two commonly used systems:

- SI (decimal) : base 10 ( $\times 1,000$ )
- IEC (binary) : base 2 ( $\times 1,024$ )

| Unit name<br>(English)                   | bit                                  | byte                                                                                                   | kilobyte                                                                                                      | megabyte                                                                                                  | gigabyte                                                                                                                  | terabyte                                                                                                                            | petabyte                                                                                                                                 |
|------------------------------------------|--------------------------------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
|                                          | <b>bit</b><br>(bit)                  | <b>octet</b><br>(o)                                                                                    | <b>kilobyte</b><br>(Ko)                                                                                       | <b>megabyte</b><br>(Mo)                                                                                   | <b>gigabyte</b><br>(Go)                                                                                                   | <b>terabyte</b><br>(To)                                                                                                             | <b>petabyte</b><br>(Po)                                                                                                                  |
| <b>Symbol</b><br>(English / French)      | <b>b</b>                             | <b>B (o)</b>                                                                                           | <b>kB (Ko)</b>                                                                                                | <b>MB (Mo)</b>                                                                                            | <b>GB (Go)</b>                                                                                                            | <b>TB (To)</b>                                                                                                                      | <b>PB (Po)</b>                                                                                                                           |
| <b>Value in bytes</b><br>(SI, decimal)   | 1/8 byte                             | 1 byte                                                                                                 | 1,000 bytes<br>( $10^3$ )                                                                                     | 1,000,000 bytes<br>( $10^6$ )                                                                             | 1,000,000,000 bytes<br>( $10^9$ )                                                                                         | 1,000,000,000,000 bytes<br>( $10^{12}$ )                                                                                            | 1,000,000,000,000,000 bytes<br>( $10^{15}$ )                                                                                             |
| <b>Value in bytes</b><br>(IEC, binary)   | 1/8 byte                             | 1 byte                                                                                                 | 1,024 bytes<br>( $2^{10}$ )                                                                                   | 1,048,576 bytes<br>( $2^{20}$ )                                                                           | 1,073,741,824 bytes<br>( $2^{30}$ )                                                                                       | 1,099,511,627,776 bytes<br>( $2^{40}$ )                                                                                             | 1,125,899,906,842,624 bytes<br>( $2^{50}$ )                                                                                              |
| <b>Concrete example</b><br>(approximate) | <div>0 1</div> A single binary digit |  A character<br>("A") |  A short text<br>(~ 1 page) |  A photo<br>(~ 2–5 MB) |  One hour<br>of HD video<br>(~ 1–2 GB) |  ~ 250,000 photos<br>or 500 hours<br>of HD video |  Data from a large<br>research facility<br>(per year) |

## Key takeaways

- 1 byte (B) = 8 bits (b)
- SI units use base 10 (1 kB = 1,000 B, 1 MB = 1,000 kB, ...) → Common in everyday use (Ko, Mo, Go, To, Po).
- IEC units use base 2 (1 KiB = 1,024 B, 1 MiB = 1,024 KiB, ...) → Common in operating systems, RAM and some software tools.
- File sizes may appear slightly different depending on the system and on rounding (e.g. a "1 GB" file may show as 0.93 GiB).

## Quick reference

| SI (decimal, base 10) | IEC (binary, base 2) |
|-----------------------|----------------------|
| 1 kB = 1,000 B        | 1 KiB = 1,024 B      |
| 1 MB = 1,000 kB       | 1 MiB = 1,024 KiB    |
| 1 GB = 1,000 MB       | 1 GiB = 1,024 MiB    |
| 1 TB = 1,000 GB       | 1 TiB = 1,024 GiB    |
| 1 PB = 1,000 TB       | 1 PiB = 1,024 TiB    |

## Which units to use?

- Use b/B for data rates (e.g. 1 Gb/s).
- Use B, kB, MB, GB, ... (or o, Ko, Mo, Go, ...) for storage and file sizes.
- Be aware of the difference between decimal and binary units.
- In most contexts (e.g. sequencing data, omics files), GB/Go and TB/To refer to decimal units (base 10).



## How a computer works

- RAM: working memory, erased when powered off
- Disk/SSD: storage, kept when powered off

| Storage / Memory | Typical speed | Typical cost per GB | Capacity  | Main use                          |
|------------------|---------------|---------------------|-----------|-----------------------------------|
| HDD              | ~100–250 MB/s | ~\$0.015–0.03/GB    | ★ ★ ★ ★ ★ | Large datasets, archives, backups |
| SATA SSD         | ~500–550 MB/s | ~\$0.05–0.10/GB     | ★ ★ ★ ★   | General storage, applications     |
| NVMe SSD         | ~3–14 GB/s    | ~\$0.06–0.15/GB     | ★ ★ ★     | Fast storage, active datasets     |
| RAM              | ~30–100+ GB/s | ~\$2–5/GB           | ★         | Data actively used by programs    |

# How a computer works

- RAM: working memory, erased when powered off
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| Omics data type                    | Typical raw data / sample |
|------------------------------------|---------------------------|
| Genotyping array / SNP chip        | 10-100 MB                 |
| Bulk RNA-seq                       | 2-10 GB                   |
| ATAC-seq                           | 2-10 GB                   |
| ChIP-seq                           | 2-10 GB                   |
| Whole-genome bisulfite seq. (WGBS) | 20-100+ GB                |
| Whole-genome sequencing (WGS)      | 30-150+ GB                |
| scRNA-seq / snRNA-seq              | 20-100+ GB                |
| scATAC-seq                         | 20-100+ GB                |
| Spatial transcriptomics            | 10-100+ GB                |
| Imaging                            | 100 GB-several TB         |
| Long-read WGS (ONT/PacBio)         | 50-300+ GB                |

- RAM: working memory, erased when powered off
- Disk/SSD: storage, kept when powered off
- **CPU: does the calculations**
- **GPU: many calculations in parallel (images, deep learning)**



- CPU: does the calculations
- GPU: many calculations in parallel (images, deep learning)



## How a computer works

- OS: manages all of the components

**Operating System (OS):**  
manages hardware resources and provides an interface between applications and hardware.

# HOW OPERATING SYSTEM WORKS

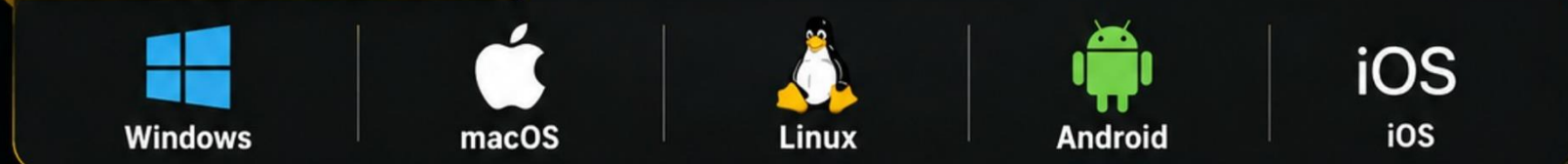
The Operating System is the **brain** of your computer. It manages everything behind the scenes!



### MAIN FUNCTIONS



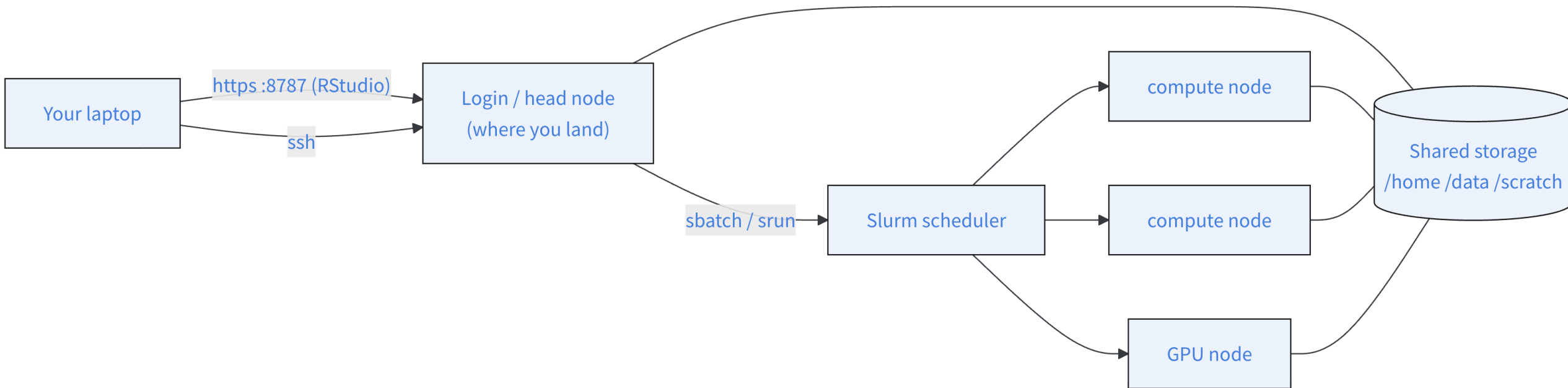
### POPULAR OPERATING SYSTEMS



- A **path** says where a file lives
- The **working directory** is where a command looks by default
- The **shell**: type instead of click, one command at a time



## From your laptop to a compute cluster



- LEGcompute: ~1000 CPU cores, 4 TB RAM, 80 TB SSD, 10 GPUs
- You share the login node; heavy jobs go through Slurm (later sessions)

- UNINE wifi or GlobalProtect VPN
- Claim your account: Google Sheet ([to this links](#))
- **RStudio Server** → Terminal tab (what we use today)
- SSH: for later, once you're comfortable
- 100 GB quota per student. No backups: it's on you.

# The FASTA format

```
>NC_045512.2 partial genome  
ATGTTTGTTTTTCTTGTTTATTGCC...
```

- > line = header, name of the sequence
- Line(s) after = the sequence itself
- The format does not tell you: DNA or protein? complete? correct? → you have to look

## Useful Databases for Functional Annotation

| Examples of useful functional annotation databases |                 |                                                                                                                                     |
|----------------------------------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Database                                           | Number of terms | Associated bioinformatics tools                                                                                                     |
| <a href="#">Gene Ontology</a> (GO)                 | 45000           | DAVID ("Database for Annotation, Visualization and Integrated Discovery"), PANTHER, <a href="#">WebGestalt</a> , ClueGO, g:Profiler |
| <a href="#">KEGG</a>                               | 24000           | <a href="#">DAVID</a> , WebGestalt, ClueGO (Cytoscape plug-in), g:Profiler                                                          |
| <a href="#">Reactome</a>                           | 21000           | Reactome, PANTHER, WebGestalt, ClueGO, <a href="#">g:Profiler</a>                                                                   |
| <a href="#">WikiPathways</a>                       | 3000            | WebGestalt ("WEB-based GEnome SeT AnaLysis Toolkit")                                                                                |
| <a href="#">PANTHER</a>                            | 180             | PANTHER, WebGestalt                                                                                                                 |
| Source : <a href="#">Bond et al. (2021)</a>        |                 |                                                                                                                                     |